# Haoran (Jensen) Yuan

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## EDUCATION

**Doctor of Philosophy in Computer Engineering** — Aug. 2026 – May 2028 (Expected)
*University of Illinois Urbana-Champaign | Advisor: Prof. Ismini Lourentzou* — Champaign, IL

**Master of Science in Computer Engineering** — Sep. 2024 – May 2026
*University of Illinois Urbana-Champaign* — Champaign, IL

**Bachelor of Science in Electrical Engineering, Minor in Computer Science** — Sep. 2020 – Jun. 2024
*University of Illinois Urbana-Champaign* — Champaign, IL

**Bachelor of Engineering in Computer Engineering** — Sep. 2020 – Jun. 2024
*Zhejiang University* — Hangzhou, China

## RESEARCH INTERESTS

Efficient AI, with a focus on making **embodied AI generalizable, scalable, and efficient**: world-action models, representation learning, visuo-tactile learning, lossless speculative decoding, agentic robotics, reinforcement learning for world models, and recursive self-improvement (RSI).

## PUBLICATIONS

** equal contribution  † project lead*

[1] **DexTacWAM: A Visuo-Tactile World-Action Model for Dexterous Manipulation**
**H. Yuan**†, Z. Wang, B. Shao, H. Lu, T. Darrell, I. Lourentzou, W. Zhan. *arXiv:2609.24976*, 2026.
[project] [paper] [code]

[2] **VTAM: Video-Tactile-Action Models for Complex Physical Interaction Beyond VLAs**
**H. Yuan***†, W. Yi*, Z. Zhang*, W. Chen*, Y. Mo, J. Yin, X. Li, X. Zeng, C. Wen, C. Lu, K. Driggs-Campbell, I. Lourentzou. *arXiv:2603.23481*, 2026.
[project] [paper]

[3] **JetSpec++: Foresight Parallel Drafting with Self Refinements**
Z. Feng*, L. Hu*, **H. Yuan**, Y.-Y. Qian, J. Yang, P. Zhao, Y. Zhu, T. Rosing, H. Zhang. *arXiv:2610.19792*, 2026.
[paper] [code]

[4] **JetSpec: Breaking the Scaling Ceiling of Speculative Decoding with Parallel Tree Drafting**
L. Hu, Z. Feng, Y. Wu, **H. Yuan**, Y. Zhao, Y.-Y. Qian, B. Wang, P. Zhao, D. Jiang, Y. Zhu, T. Rosing, H. Zhang. *arXiv:2606.18394*, 2026.
[project] [paper] [code]

## RESEARCH EXPERIENCE

**DexTacWAM: Visuo-Tactile World-Action Model (Project Lead, BAIR @ UC Berkeley)** | *World Model, Tactile*

- Led a visuo-tactile **World-Action Model** that adapts a pretrained video DiT to 10 fingertip tactile sensors on a 22-DoF bimanual robot via **modality-split flow matching**, modeling contact that cameras miss under occlusion.
- Designed a finger- and pose-aware **tactile compressor** fusing 10 fingertip latents into 2 hand views, retaining **89.4%** of contact recall while cutting DiT tokens by **3.67**× and training time by **2.26**×.
- Achieved **70.6%** avg. success on 6 real-robot tasks vs. **38.0%** for the best baseline (RDP); featured by BAIR.

**VTAM: Video–Tactile Action Model (Project Lead, PLAN Lab @ UIUC)** | *PyTorch, World Model, Multimodal Learning*

- Led end-to-end development (2k+ real-world trajectories), adapting a pretrained video transformer to touch **without tactile pretraining**; reached **90% success** on chip pick-and-place, plate wiping and cucumber peeling.
- Designed a gated cross-attention mechanism to suppress cross-modal contamination between tactile and visual tokens, reducing training loss spikes by **42%** and speeding up convergence by **1.7**×.
- Added a force-aware diffusion regularizer to balance tactile–vision attention: **+80%** success over vision-only.

**JetSpec++: Foresight Parallel Drafting with Self Refinements (Core Contributor, Hao AI Lab @ UCSD)**

- Developed a **causal correction for parallel drafting** that conditions each drafted token on its **full branch history** rather than the preceding token alone, pre-drafting the next block during verification.
- At matched parameter count, **outperformed DeepSeek's DSpark**, with the margin widening at draft depths beyond those used in training; raised average acceptance length to **8.94** (vs. 6.89).
- Achieved up to **7.45**× lossless speedup over AR and **1.54**× over DSpark on Qwen3-8B (**1,361 tok/s**, GB200).

**JetSpec: Parallel Tree Drafting for Speculative Decoding (Core Contributor, Hao AI Lab @ UCSD)**

- Co-developed a **causal parallel draft head** that emits a complete scored speculative tree in **one forward pass** while preserving the target model's autoregressive factorization along every branch.
- Integrated with a standalone engine using FlashAttention **tree-attention kernels** (Triton, CuTe DSL) and vLLM.
- Achieved up to **9.64**× lossless speedup over AR on Qwen3-8B (MATH-500), beating DDTree (8.78×) and EAGLE-3.

## WORK EXPERIENCE

**Snowflake** | Software Engineering Intern — Menlo Park, CA | **May 2025 – Aug. 2025**

- Designed and implemented a **predictive autoscaler** for Snowflake GS clusters, combining Gradient Boosting, ARIMA and XGBoost models trained on historical telemetry from the Snowhouse database.
- Achieved **14.3% MAPE** and **92.5% coverage** at the 95% confidence interval for **24-hour** CPU forecasts using under one month of history, with end-to-end training and prediction in <5 minutes per deployment.
- Deployed on the internal background job system to adjust instance provisioning, increasing average **CPU utilization by >20%** and reducing overprovisioning in simulated datacenter environments.

## TECHNICAL SKILLS

**Languages:** Python, C/C++, Rust, Java, SQL, SystemVerilog.
**Frameworks:** PyTorch, CUDA, Triton, SGLang, vLLM, Isaac Gym, Isaac Sim, MuJoCo, Flask, SQLAlchemy, OpenCV.
**Databases & Tools:** MySQL, PostgreSQL, Redis, Docker, AWS, GCP, Git, Linux, Shell.